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(54) **VERIFYING AUTHORITY IN DISTRIBUTED SYSTEMS**

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(52) **U.S. Cl.**
CPC **H04L 9/3265** (2013.01); **H04L 9/3271** (2013.01)

(58) **Field of Classification Search**
CPC H04L 9/3265; H04L 9/3271
See application file for complete search history.

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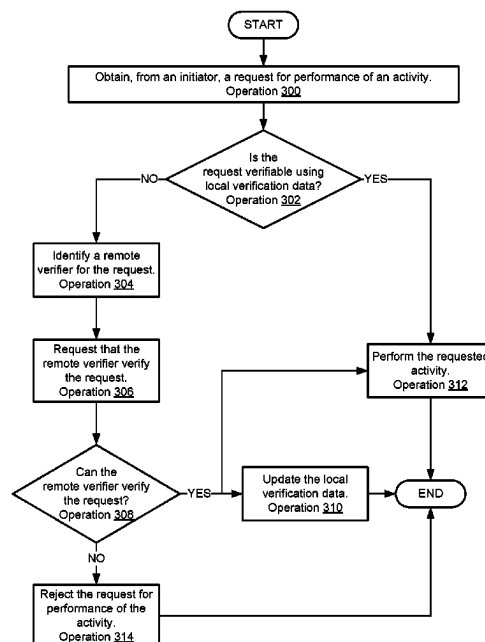
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(57) **ABSTRACT**

Methods and systems for securing endpoint devices are disclosed. To secure the endpoint devices, a flexible system for verifying authority may be used. The flexible system may allow data processing system in resource constrained environment to invoke the functionalities of other data processing systems. The flexible system may reduce the resources expended for invoking functionalities of other data processing systems by not requiring that requests for the invocation of the functionalities include sufficient verification for the other data processing systems to verify that the requestors have sufficient authority to invoke the functionalities of the other data processing systems.

20 Claims, 6 Drawing Sheets



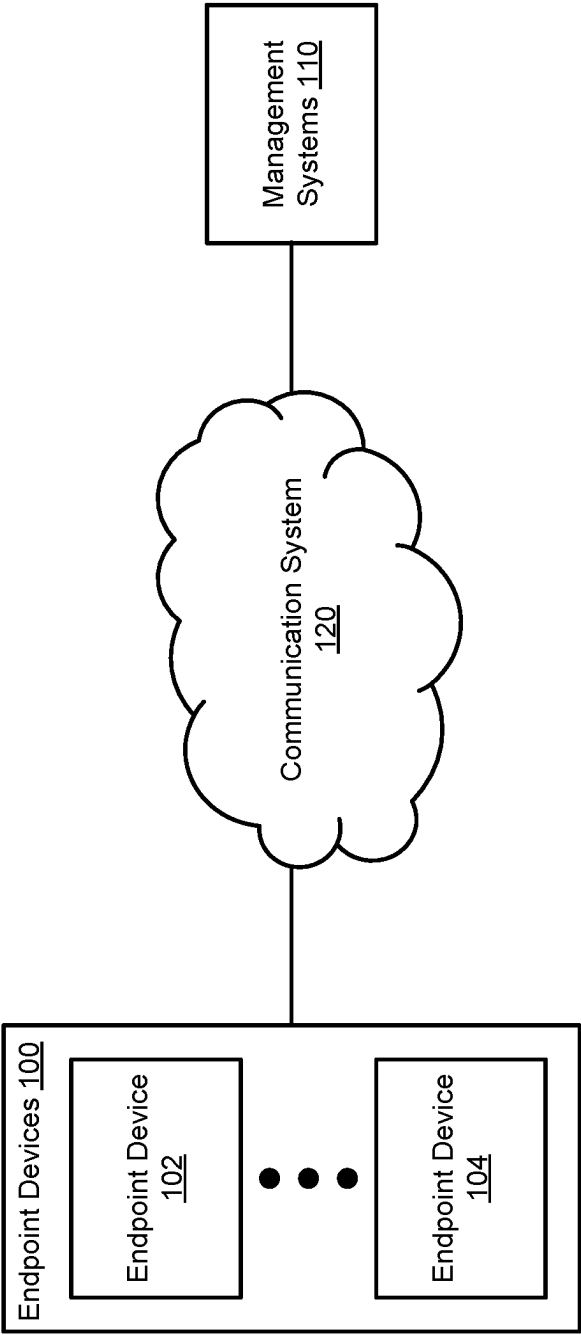


FIG. 1

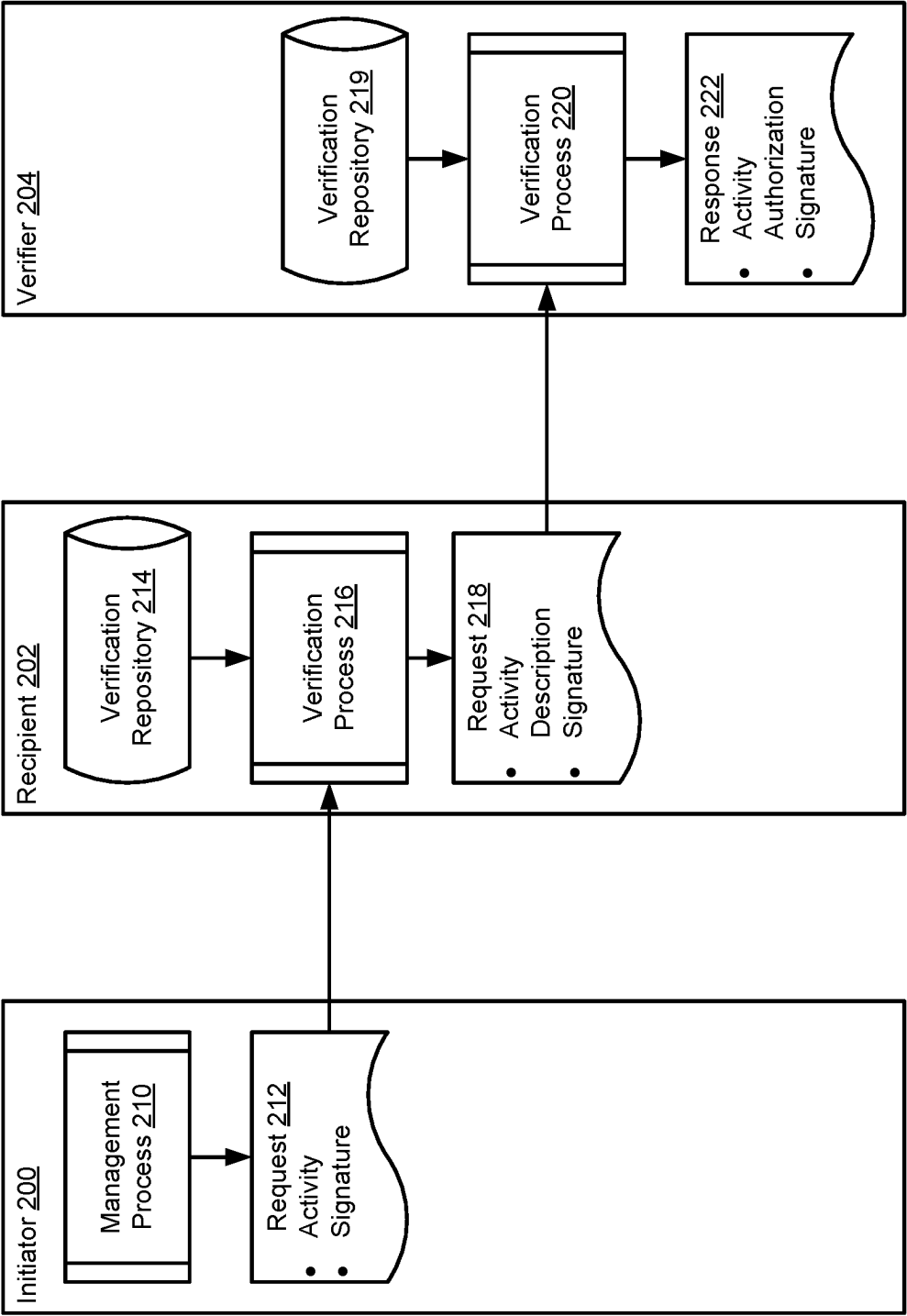


FIG. 2A

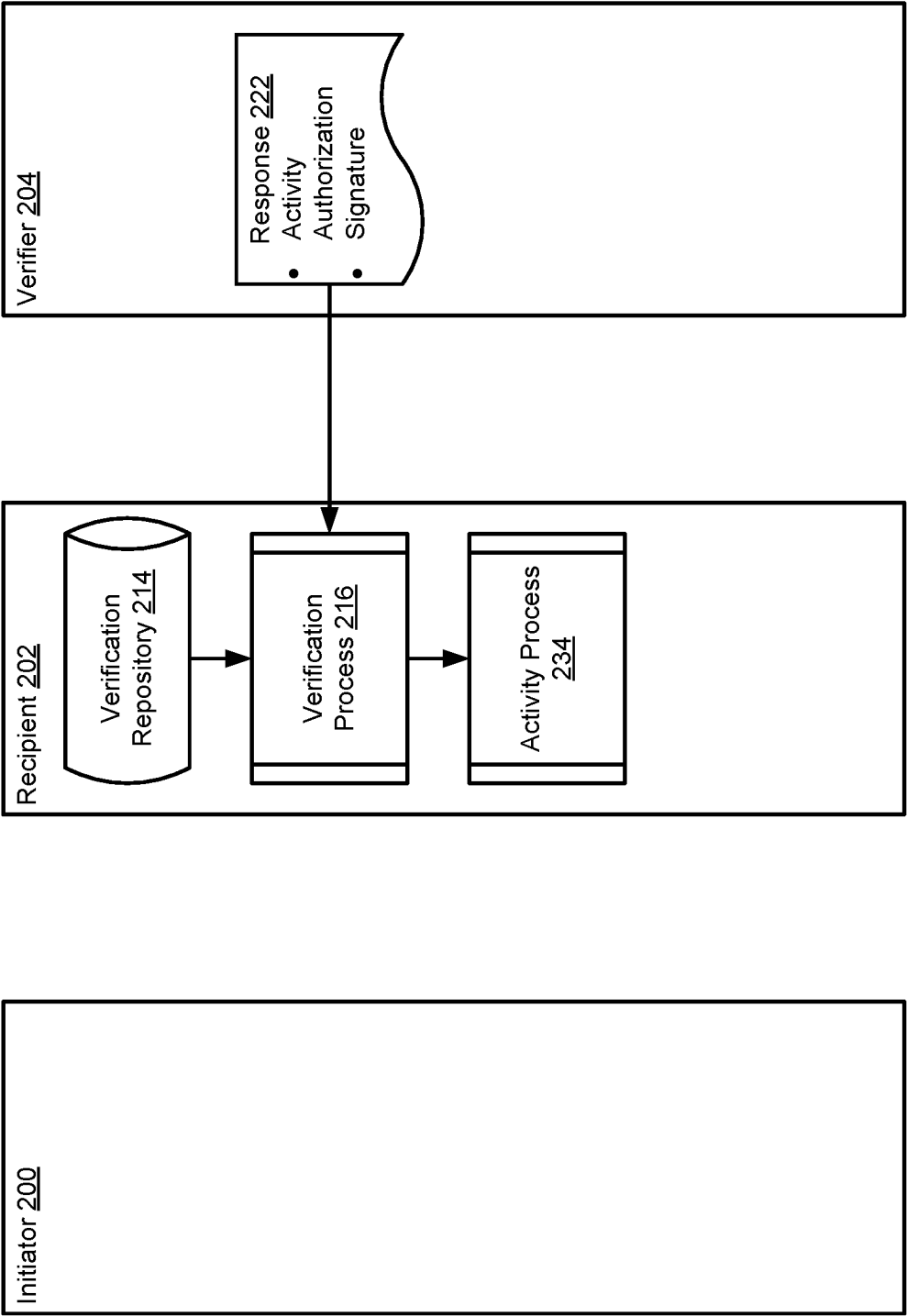


FIG. 2B

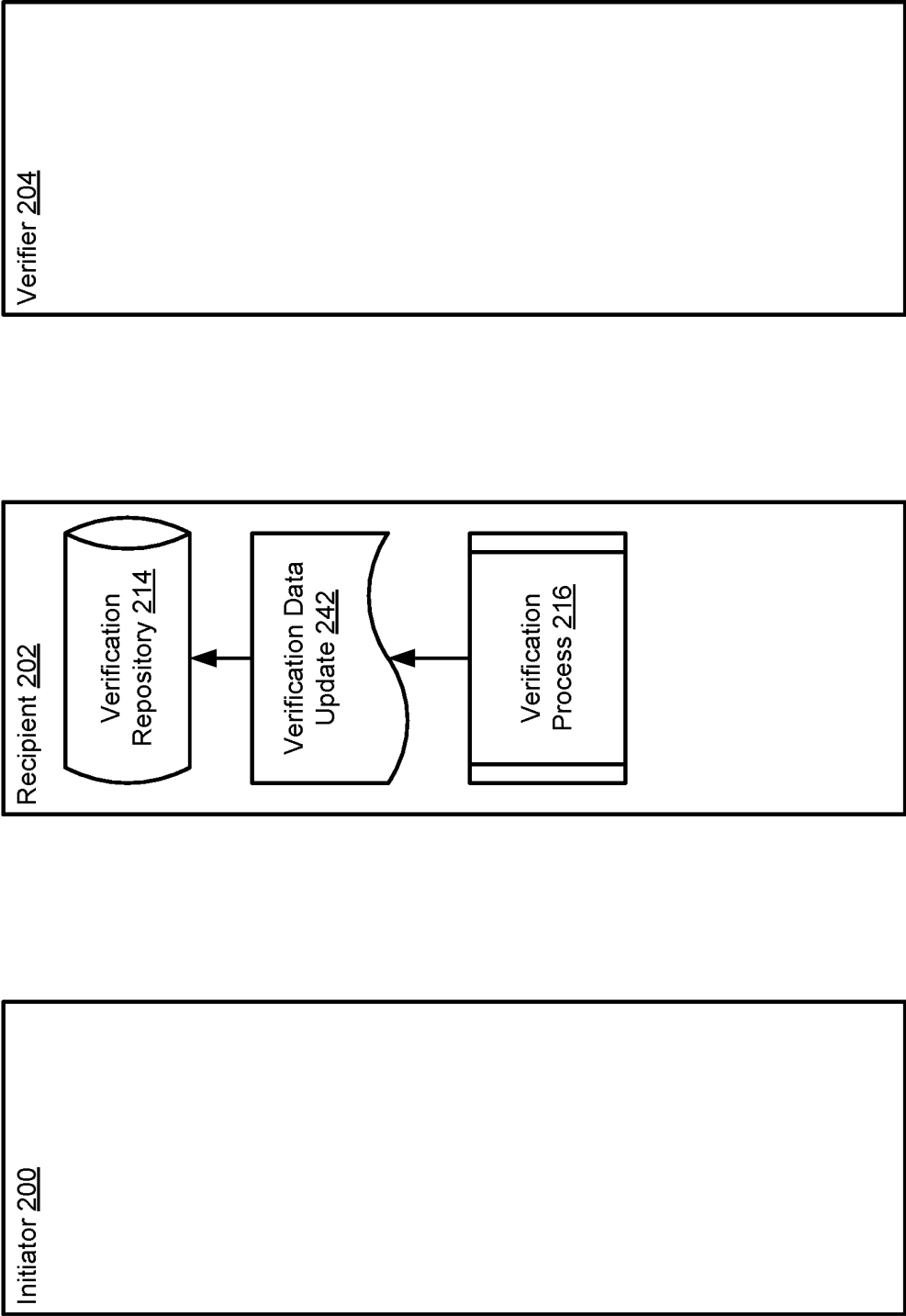


FIG. 2C

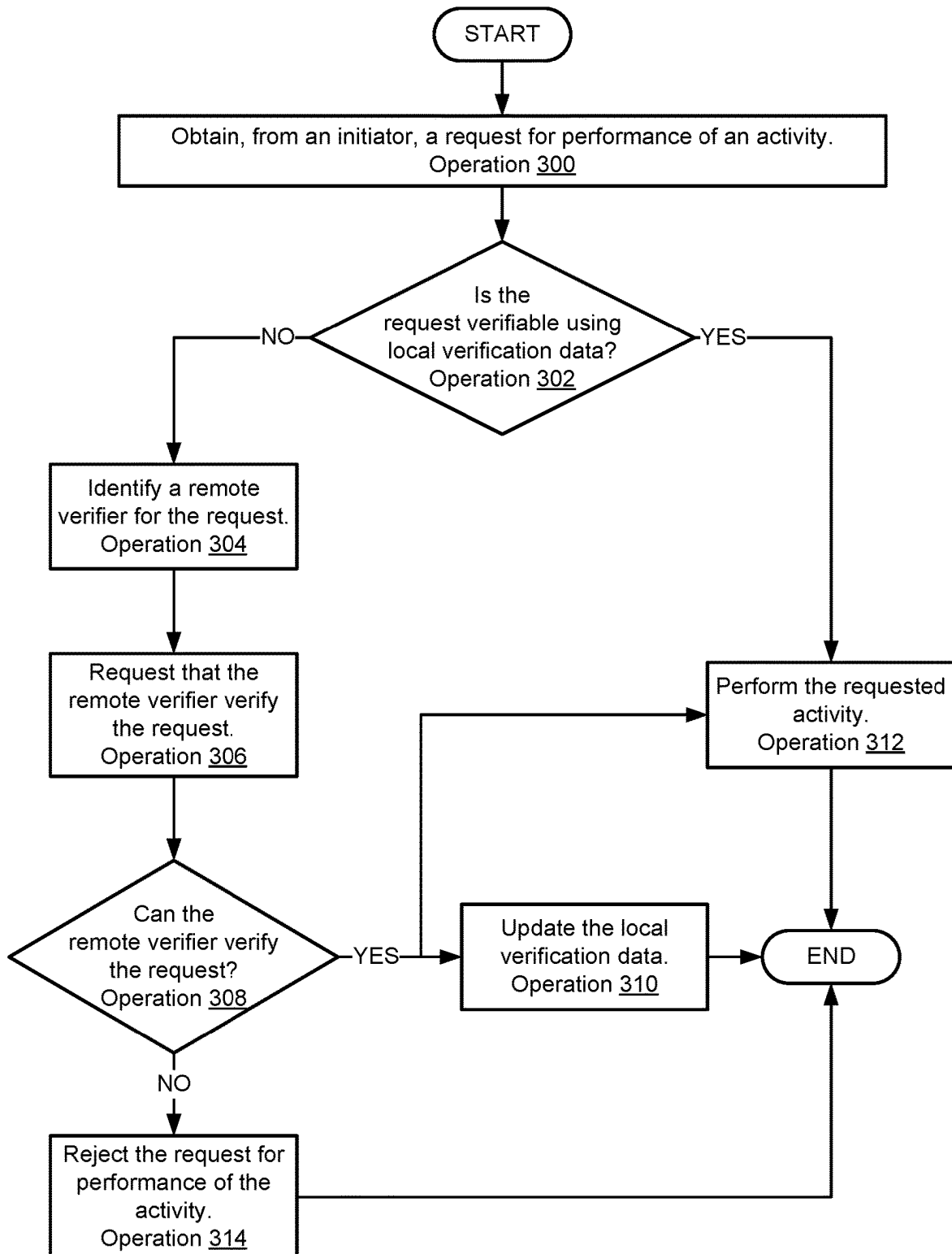


FIG. 3

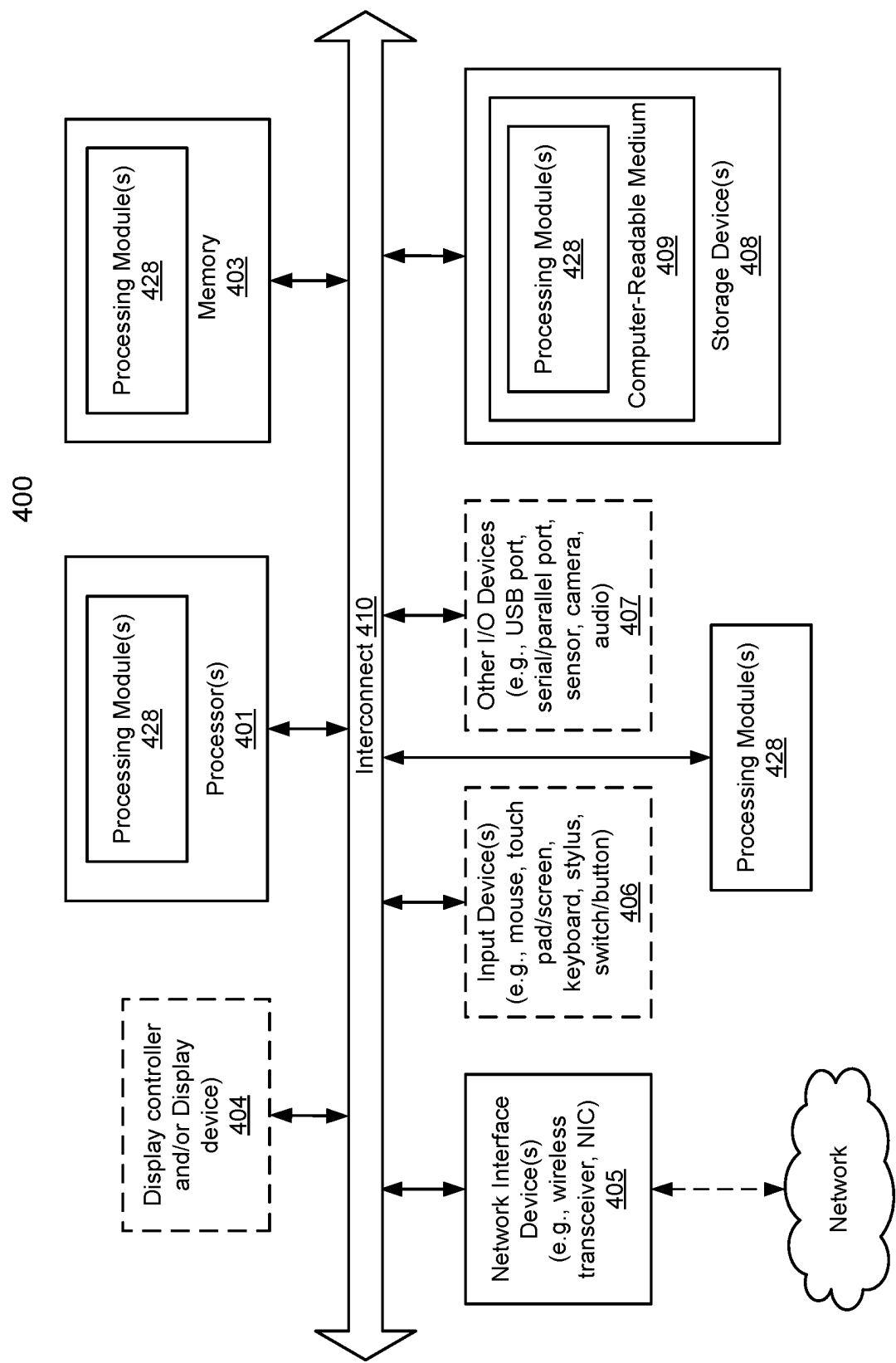


FIG. 4

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VERIFYING AUTHORITY IN DISTRIBUTED SYSTEMS

FIELD

Embodiments disclosed herein relate generally to security. More particularly, embodiments disclosed herein relate to securing devices.

BACKGROUND

Computing devices may provide computer-implemented services. The computer-implemented services may be used by users of the computing devices and/or devices operably connected to the computing devices. The computer-implemented services may be performed with hardware components such as processors, memory modules, storage devices, and communication devices. The operation of these components and the components of other devices may impact the performance of the computer-implemented services.

BRIEF DESCRIPTION OF THE DRAWINGS

Embodiments disclosed herein are illustrated by way of example and not limitation in the figures of the accompanying drawings in which like references indicate similar elements.

FIG. 1 shows a block diagram illustrating a system in accordance with an embodiment.

FIGS. 2A-2C show data flow diagrams in accordance with an embodiment.

FIG. 3 shows a flow diagram illustrating method in accordance with an embodiment.

FIG. 4 shows a block diagram illustrating a data processing system in accordance with an embodiment.

DETAILED DESCRIPTION

Various embodiments will be described with reference to details discussed below, and the accompanying drawings will illustrate the various embodiments. The following description and drawings are illustrative and are not to be construed as limiting. Numerous specific details are described to provide a thorough understanding of various embodiments. However, in certain instances, well-known or conventional details are not described in order to provide a concise discussion of embodiments disclosed herein.

Reference in the specification to “one embodiment” or “an embodiment” means that a particular feature, structure, or characteristic described in conjunction with the embodiment can be included in at least one embodiment. The appearances of the phrases “in one embodiment” and “an embodiment” in various places in the specification do not necessarily all refer to the same embodiment.

References to an “operable connection” or “operably connected” means that a particular device is able to communicate with one or more other devices. The devices themselves may be directly connected to one another or may be indirectly connected to one another through any number of intermediary devices, such as in a network topology.

In general, embodiments disclosed herein relate to methods and systems for securing operation of endpoint devices using a flexible approach. The flexible approach may allow for functionalities of a data processing systems to be invoked in resource constrained environments.

For example, a first data processing system (e.g., an “initiator”) may need to invoke functionality of a second

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endpoint device (e.g., a “recipient”). The initiator may do so by sending a request to the recipient. To establish that the initiator has been delegated authority to invoke the functionality, the recipient may need to have access to verification information that is sufficient to cryptographically verify that the initiator has been delegated the authority. However, due to network communication limitations between the initiator and the recipient and/or for lack of having access to certain verification information, initiator may not be able to provide sufficient verification information to the recipient to enable recipient to verify authority of the initiator.

Rather than rejecting the request for lack of ability to verify the authority of the initiator, the recipient may take proactive action to determine whether the initiator has the necessary authority. To do so, the recipient may rely on a third data processing system (e.g., a “verifier”) to verify the authority of the initiator. The verifier may have access to additional verification information which the recipient did not have access to while trying to verify the authority of the initiator.

By doing so, embodiments disclosed herein may allow data processing systems located in resource constrained environment (e.g., lacking bandwidth and/or access to verification information) to invoke the functionalities of other data processing systems. By doing so, the data processing systems within the resource constrained environment may be more likely to be able to provide desired computer implemented services.

In an embodiment, a method for managing operation of devices in a distributed system is provided. The method may include obtaining, by a recipient and from an initiator, a request for performance of an activity; making a first determination regarding whether the request is verifiable using local verification data maintained by the recipient; in a first instance of the first determination where the request is not verifiable: identifying a remote verifier for the request; requesting that the remote verifier attempt to verify the request; making a second determination regarding whether the remote verifier successfully verified the request; and in a first instance of the second determination where the remote verifier successfully verified the request; performing, by the initiator, the activity.

The method may also include, in a second instance of the first determination where the request is verifiable: performing, by the initiator, the activity.

The method may also include, in a second instance of the second determination where the remote cannot verify the request: rejecting the request for performance of the activity.

Making the first determination may include identifying a signature from the request; and identifying that the local verification data lacks a certificate chain delegating authority for the activity to an entity that establish the certificate and that is usable to authenticate the signature with respect to a root of trust for the recipient.

Requesting that the remote verifier attempt to verify the request may include generating a challenge for the remote verifier, the challenge requiring cryptographic data usable to establish trust in the entity; and sending the challenge to the remote verifier.

Making the second determination may include obtaining a challenge response to the challenge; and in an instance of the obtaining when the challenge response comprises a signature of a trusted entity and an endorsement of authority of the entity to initiate the activity: concluding that the remote verifier verified the request.

Requesting that the remote verifier attempt to verify the request further may include identifying that the remote

verifier is unreachable. The challenge may be sent to the remote verifier after the remote verifier becomes reachable.

The request may not include verification data for the request, the verification data being usable to verify that the initiator has authority to initiate the activity by the recipient.

The method may also include obtaining, by the recipient and from a second initiator, a second request for performance of an activity, the second request comprising second verification data for the second request, the second verification data being usable to verify that the second initiator has authority to initiate the activity by the recipient.

The initiator may be operably connected to the recipient via a limited channel (e.g., a limited communication channel, a limited communication link, etc.), the limited channel preventing the verification data from being part of the request.

In an embodiment, a non-transitory media is provided. The non-transitory media may include instructions that when executed by a processor cause the computer-implemented method to be performed.

In an embodiment, a data processing system is provided. The data processing system may include the non-transitory media and a processor, and may perform the method when the computer instructions are executed by the processor.

Turning to FIG. 1, a block diagram illustrating a system in accordance with an embodiment is shown. The system shown in FIG. 1 may provide computer-implemented services. The computer implemented services may include any type and quantity of computer implemented services. For example, the computer implemented services may include data storage services, instant messaging services, database services, and/or any other type of service that may be implemented with a computing device.

To provide the computer implemented services, the system may include endpoint devices **100**. Each endpoint device (e.g., **102**, **104**) may provide similar and/or different computer implemented services, and may provide the computer implemented services independently and/or in cooperation with other endpoint devices.

To provide the services, each of endpoint devices **100** may utilize functionalities of other endpoint devices. For example, a first endpoint device (e.g., **102**) may request that a second endpoint device (e.g., **104**) perform a certain activity, series of activities, perform certain actions, etc. When so performed, the activity of the second endpoint device may allow the first endpoint device to accomplish a goal, objective, etc.

To invoke the functionalities of other endpoint devices, a first endpoint device may send a request to a second endpoint device. The request may indicate the type of activity to perform. When a second endpoint device obtains the request, the second endpoint device may perform the type of the activity thereby contributing to a goal, objective, etc. of the first endpoint device.

To reduce the potential for malicious activity to impact operation of endpoint device **100**, endpoint devices **100** may implement an authentication framework for requests. The authentication framework may require that authority for a first endpoint device to invoke a functionality of a second endpoint device be verified prior to granting requests that invoke the functionality. For example, if a first endpoint device attempts to invoke data storage functionality of a second endpoint device, the second endpoint device may verify that the first endpoint device has authority to invoke data storage functionality. Consequently, the authentication framework may limit the ability of malicious entities to invoke functionalities of endpoint devices.

To verify authority, an endpoint device may attempt to establish chains of delegation of authority from a root of trust for the endpoint device to an entity attempting to invoke a functionality of the endpoint device. The chains of delegation may be established using, for example, (i) certificates that delegate authority, (ii) attestations that verify authority, and/or other types of cryptographically verifiable data structures.

For example, a data processing system may maintain a public key usable to verify signatures (e.g., made using a corresponding private key) from the root of trust. To establish the delegation, the data processing system may identify a chain of certificates that establishes a delegation of authority for invocation of a functionality from the root of trust to an entity (e.g., an "initiator") trying to invoke the functionality. The data processing system may do so by sequentially identifying certificates that include (i) delegations of the authority between entities, and (ii) that establish a cryptographically verifiable chain from the root of trust to the initiator. The cryptographically verifiable chain may include certificates that include public keys of entities to which authority is being delegated and that are signed using private keys of the entities making the delegation. In this manner, the certificates may establish a cryptographically verifiable chain of delegations from the root of trust to the initiator.

In another example, the chain may include an attestation of a trusted entity. That attestation may attest to the authority of the initiator, and may be signed using a private key of the attestor. If the attestor is trusted by the data processing system, then the attestation may serve as verification of the authority of the initiator.

To facilitate verification of authority of the initiator, the request may include certificates, attestations, and/or other types of cryptographically verifiable data structures. By including these cryptographically verifiable data structures, the recipient of the request may have access to all of the information necessary to verify the request.

However, if the request does not include these cryptographically verifiable data structures and the recipient does not have access to them, then the recipient may be unable to verify the request. For example, if two endpoint devices are operably connected to one another via a limited connection (e.g., low bandwidth, low capacity), then requests that include the cryptographically verifiable data may not be transmitted. In another example, if an initiating endpoint device (e.g., an initiator) lacks access to some cryptographically verifiable data structures necessary to verify the delegation of authority, then the requests may not be self-verifying to the recipient endpoint device.

If an endpoint device is unable to transit self-verifying requests to other devices, then the endpoint device may be unable to invoke the functionalities of other endpoint devices. Accordingly, the endpoint device may be unable to meet its objective or other goals for lack of ability to utilize functionalities of the distributed system.

In general, embodiments disclosed herein may provide methods, systems, and/or devices for securing functionalities of endpoint devices in distributed systems. To secure the functionality of the endpoint devices, the endpoint devices may implement a flexible verification model. The flexible verification model may allow requests from other devices to be verified even when the requests are not self-verifiable and/or verifiable using local verification information.

To verify such requests, the endpoint devices may, when the requests are received, rely on other devices to provide information usable to verify that the requests are issued by entities with sufficient delegated authority. The endpoint

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devices may generate and send challenges to these other devices. The challenges may require that the other devices provide information in a verifiable format that allows for the requests to be verified.

For example, a first endpoint devices may issue a challenge to management system **110**. In response, management system **110** may provide a verifiable response with attestations, certificates, and/or other types of cryptographically verifiable data structures. These data structures may include information that establishes that a requestor has been delegated authority to invoke functionalities of the first endpoint device.

By doing so, functionalities of endpoint devices may be invoked even though sufficient information to verify requests for the functionalities is unavailable to a recipient device. Consequently, initiating devices that may be unable to provide self-verifying requests may be able to invoke the functionalities of endpoint devices. Accordingly, the endpoint devices may be used in more challenging environments that may preclude transmission of self-verifying requests.

For example, if a first endpoint device is operably connected to a second endpoint device via a communication link (e.g., bandwidth limited, time limited, etc.) that prevents transmission of a self-verifiable request, the first endpoint devices may send a request without verification information. The recipient device may automatically take actions to obtain information to verify the request rather than merely refusing it for lack of ability to verify the request initially. If the recipient is similarly limited in connectivity, the recipient may wait until connectivity improves to verify the request.

To provide the above noted functionality, the system of FIG. **1** may include endpoint devices **100** and management system **110**. Each of these components is discussed below.

Endpoint devices **100** may provide computer implemented services that may be secured using a flexible verification framework. The flexible verification framework may allow for both self-verifying requests and non-self-verifying requests to be verified (or at least attempted to be verified). Once verified, the activity indicated by the requests may be performed. Similarly, endpoint devices may use a flexible request framework. The flexible request framework may facilitate dynamic generation of self-verifying requests and non-self-verifying requests depending on the conditions to which an endpoint device is exposed. Refer to FIGS. **2A-2C** for additional details regarding the operation of endpoint devices **100**.

Management system **110** may participate in verification of requests. For example, endpoint devices **100** may treat management system **110** as a remote source of verification information for requests. Management system **110** may participate in the flexible verification framework by (i) obtaining challenges, based on received requests, from endpoint devices **100**, (ii) analyze the challenges to formulate responses that include information usable by endpoint devices **100** to verify the requests, and (ii) provide the responses to endpoint devices **100**. The management system **110** may include various types of cryptographically verifiable data structures including, for example, certificates defining delegations of authority, and include functionality to generate cryptographically verifiable data structures such as attestations. Refer to FIGS. **2A-2C** for additional details regarding the operation of management system **110**.

When providing their functionality, any of (and/or components thereof) endpoint devices **100** and/or management system **110** may perform all, or a portion, of the method illustrated in FIG. **3**.

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Any of (and/or components thereof) endpoint devices **100** and management system **110** may be implemented using a computing device (also referred to as a data processing system) such as a host or a server, a personal computer (e.g., desktops, laptops, and tablets), a “thin” client, a personal digital assistant (PDA), a Web enabled appliance, a mobile phone (e.g., Smartphone), an embedded system, local controllers, an edge node, and/or any other type of data processing device or system. For additional details regarding computing devices, refer to FIG. **4**.

Any of the components illustrated in FIG. **1** may be operably connected to each other (and/or components not illustrated) with communication system **120**. In an embodiment, communication system **120** includes one or more networks that facilitate communication between any number of components. The networks may include wired networks and/or wireless networks (e.g., and/or the Internet). The networks may operate in accordance with any number and types of communication protocols (e.g., such as the internet protocol).

In an embodiment, the system of FIG. **1** comprises another communication system that operably connects various endpoint devices **100** to one another. The communication system may be implemented using limited connections.

For example, the limited connections may include bandwidth limited connections (e.g., low bandwidth), payload size limited connections (e.g., radio frequency identification tag based communication systems which may be capable of storing specific sized payloads and transmitting that size of payload), time varying connections which may limit when data may be sent, and/or other types of limited connections. Communication system **120** may include some similar types of connections. Consequently, any of endpoint devices **100** and management systems **110** may be limited in communication capability. Consequently, any of endpoint devices **100** may elect to send requests that may not be self-verifying and/or verifiable using information locally available to a recipient.

While illustrated in FIG. **1** as including a limited number of specific components, a system in accordance with an embodiment may include fewer, additional, and/or different components than those components illustrated therein.

To further clarify embodiments disclosed herein, data flow diagrams in accordance with an embodiment are shown in FIGS. **2A-2C**. In these diagrams, flows of data and processing of data are illustrated using different sets of shapes. A first set of shapes (e.g., **212**, **216**, etc.) is used to represent data structure, a second set of shapes (e.g., **210**, **216**, etc.) is used to represent processes performed using and/or that generate data, and a third set of shapes (e.g., **214**) is used to represent large scale data structures such as databases.

Turning to FIG. **2A**, a first data flow diagram in accordance with an embodiment is shown. The first data flow diagram may illustrate data used in and data processing performed in invocation of functionality of data processing systems, and verification of requests for the invocation of the functionality.

To invoke a functionality of another device, initiator **200** (e.g., a first data processing system) may perform management process **210**. Management process **210** may manage interactions with other devices. For example, management process **210** may be performed by management entities such as operating systems, drivers, etc. that utilize various hardware resources of initiator **200**.

During management process **210**, requests from applications (not shown) or other entities hosted by initiator **200**

may be obtained. These requests may be for functionalities of other devices. For example, a request may be generated by an application (e.g., a database application). The request may be for another instance of the application to perform a data processing action to bring data managed by the other application into a consistent state with the data of the application.

When management process **210** obtains the request for activity to be performed by another entity, management process **210** may identify a corresponding recipient for the request and sign the request thereby obtaining request **212**. Once obtained, request **212** may be sent to recipient **202** (e.g., another endpoint device or other type of data processing system).

When received by recipient **202**, request **212** may be processed via verification process **216**. During verification process **216**, request **212** may be analyzed to ascertain whether sufficient verification information is available to confirm that initiator **200** has authority to invoke performance of the requested activity by recipient **202**. Verification information may be included as part of or with request **212**, and may be locally available in verification repository **214**. For example, certificates, attestations, and/or other types of cryptographically verifiable data structures that delegate authority may be available from these sources and to verification process **216**. Verification process **216** may attempt to use these data structures to establish a chain of delegation for the activity to initiator **200**.

However, these sources of verification information may not include sufficient verification information to establish that delegation of authority over recipient **202** for the requested activity. Verification process **216** may analyze the available verification information (e.g., included with request, from local sources) to attempt to ascertain whether the delegation can be established. If the delegation cannot be established using the available verification information, then verification process **216** may (i) select a verifier, (ii) generate request **218**, and (iii) send request **218** to the verifier (e.g., **204**).

Request **218** may include, for example, a description of the activity to be performed, information regarding the initiator (e.g., **200**), and/or other information regarding the requested activity/initiator. The information regarding the initiator may include the signature and/or payload from request **212**. Consequently, when received by verifier **204**, verifier **204** may perform verification process **220**, which may be similar to verification process **216** but performed using verification information available to verifier **204** such as that included in verification repository **219** and/or any verification information forwarded by recipient **202** (which may include verification information available to it).

Verifier **204** may be selected by verification process **216** based on, for example, (i) lookup tables that associate different types of requests with different verifiers, (ii) lookup tables that associate different initiators with different verifiers, (iii) lookup tables that associate different activities with different verifiers, and/or other sources of information.

Any of the lookup tables may associate different types of requests/initiators/activities with prioritized lists of different initiators. If a highest priority verifier is unable to verify request **212**, then recipient **202** may sequentially attempt to verify requests **212** by sending request **218** to progressively lower priority verifiers in a priority list.

While described with respect to different lookup tables, it will be appreciated that the different combinations of types of requests/initiators/activities (and/or other types of infor-

mation regarding initiators, recipients, and requests for activities) may be associated with different priority lists.

When received by verifier **204**, request **218** may be processed by verification process **220** to obtain response **222**. Response **222** may (i) indicate whether the authority is delegated to recipient **202**, and (ii) be cryptographically verifiable (e.g., by including a signature). To indicate whether the authority is delegated, response **222** may include certificates that allow recipient **202** to establish a certificate chain to verify that initiator **200** has been delegated sufficient authority for the activity, attestations that sufficient authority has been delegated to initiator **200** for the activity (in which the cryptographically verifiable signature may be part of the attestation, as opposed to being separate from it), and/or other types of verifiable data structures usable by recipient **202** to verify request **212**.

Turning to FIG. 2B, a second data flow diagram in accordance with an embodiment is shown. The second data flow diagram may illustrate data used in and data processing performed in invocation of functionality of data processing systems, and verification of requests for the invocation of the functionality.

Continuing with the discussion from FIG. 2A, once obtained, response **222** may be sent to recipient **202**. When received by recipient **202**, verification process **216** may continue the process of attempting to verify request **212**, but using the information included in response **222**, such as certificates, attestations, etc.

If request **212** is successfully verified, activity process **234** may be initiated. During activity process **234**, the activity requested in request **212** may be performed.

Turning to FIG. 2C, a third data flow diagram in accordance with an embodiment is shown. The third data flow diagram may illustrate data used in and data processing performed in invocation of functionality of data processing systems, and verification of requests for the invocation of the functionality.

Continuing with the discussion from FIG. 2B, in addition to processing response **222** for verification purposes, verification process **216** may also generate any number of verification data updates (e.g., **242**). The verification updates may include any newly obtained verification information from response **222**. The verification updates may be used to update verification repository **214** (e.g., by adding additional verification data to it). Consequently, recipient **202** may be more likely to be able to verify requests in the future.

Thus, using the flows illustrated in FIGS. 2A-2C, a system in accordance may enable initiators that may be resource constrained to invoke the functionality of recipients even if the requests for invoking the functions and the local available information verification information are insufficient to verify the requests.

As discussed above, the components of FIG. 1 may perform various methods to manage the operation of and/or interactions between endpoint devices. FIG. 3 illustrates a method that may be performed by the components of the system of FIG. 1. In the diagram discussed below and shown in FIG. 3, any of the operations may be repeated, performed in different orders, and/or performed in parallel with or in a partially overlapping in time manner with other operations.

Turning to FIG. 3, a flow diagram illustrating a method for managing secure operation of endpoint devices in accordance with an embodiment is shown. The method may be performed by any of endpoint devices **100**, management system **110**, and/or other components of the system shown in FIG. 1.

Prior to operation **300**, an entity hosted by a data processing system (e.g., an initiator) may identify that a functionality of another data processing system (e.g., a requestor) needs to be invoked to accomplish a goal.

At operation **300**, a request for performance of an activity is obtained from an initiator. The request may be obtained by receiving it in a message, by reading it from storage, and/or via other methods. The message may be received, for example, via a limited communication channel. The limited communication channel may limit the amount of data that may be sent. Consequently, the message may not include verification data usable to verify the request.

At operation **302**, a determination is made regarding whether the request is verifiable using local verification data. The determination may be made by attempting to establish a chain of delegations of authority for a requested activity to the initiator. If the chain cannot be established, then it may be determined that the requests is not verifiable using local verification data.

If the request is not verifiable, then the method may proceed to operation **304**. Otherwise the method may proceed to operation **312**.

At operation **304**, a remote verifier for the request is identified. The remote verifier may be identified based on the initiator, the recipient of the request, and/or various information included in the request. For example, the aforementioned information may be used to perform a lookup which may return an identity of the remote verifier.

At operation **306**, a request for the remote verifier to verify the request may be made. The request may be made by generating a request and sending the request to the remote verifier. The request may challenge the remote verifier to verify that the initiator has authority to cause the recipient to perform the requested activity.

At operation **308**, a determination may be made regarding whether the remote verifier is able to verify the request. The determination may be made based on a response to the request. The response may be made by the remote verifier, and may include information indicating whether the request is verifiable. If verifiable, the response may include attestations, certificates, and/or other cryptographic information usable to cryptographically verify that the initiator has authority to initiate the activity by the requestor.

If the remote verifier can verify the request from the initiator, then the method may proceed to operations **310** and **312**. Otherwise the method may proceed to operation **314**.

At operation **310**, the local verification data is updated. The local verification data may be updated by adding the certificates, attestations, etc. from the response from the remote verifier to existing local verification data.

At operation **312**, the requested activity is performed. The requested activity may be performed by initiating performance of one or more actions by the recipient.

The method may end following operations **310** and/or **312**.

Returning to operation **308**, the method may proceed to operation **314** when the remote verifier is unable to verify the request from the recipient.

At operation **314**, the request (e.g., from the initiator) for performance of the activity by the recipient is rejected by the recipient. The request may be rejected by discarding the request, notifying the initiator that the request is rejected, and/or performing other actions to manage rejected requests.

The method may end following operation **314**.

Thus, using the method illustrated in FIG. 3, an initiator may be able to invoke functionality of a recipient even if the

recipient is initially unable to verify authority of the initiator. Consequently, the initiator may be used in more challenging connectivity environments.

Any of the components illustrated in FIGS. 1-2C may be implemented with one or more computing devices. Turning to FIG. 4, a block diagram illustrating an example of a data processing system (e.g., a computing device) in accordance with an embodiment is shown. For example, system **400** may represent any of data processing systems described above performing any of the processes or methods described above. System **400** can include many different components. These components can be implemented as integrated circuits (ICs), portions thereof, discrete electronic devices, or other modules adapted to a circuit board such as a motherboard or add-in card of the computer system, or as components otherwise incorporated within a chassis of the computer system. Note also that system **400** is intended to show a high level view of many components of the computer system. However, it is to be understood that additional components may be present in certain implementations and furthermore, different arrangement of the components shown may occur in other implementations. System **400** may represent a desktop, a laptop, a tablet, a server, a mobile phone, a media player, a personal digital assistant (PDA), a personal communicator, a gaming device, a network router or hub, a wireless access point (AP) or repeater, a set-top box, or a combination thereof. Further, while only a single machine or system is illustrated, the term "machine" or "system" shall also be taken to include any collection of machines or systems that individually or jointly execute a set (or multiple sets) of instructions to perform any one or more of the methodologies discussed herein.

In one embodiment, system **400** includes processor **401**, memory **403**, and devices **405-407** via a bus or an interconnect **410**. Processor **401** may represent a single processor or multiple processors with a single processor core or multiple processor cores included therein. Processor **401** may represent one or more general-purpose processors such as a microprocessor, a central processing unit (CPU), or the like. More particularly, processor **401** may be a complex instruction set computing (CISC) microprocessor, reduced instruction set computing (RISC) microprocessor, very long instruction word (VLIW) microprocessor, or processor implementing other instruction sets, or processors implementing a combination of instruction sets. Processor **401** may also be one or more special-purpose processors such as an application specific integrated circuit (ASIC), a cellular or baseband processor, a field programmable gate array (FPGA), a digital signal processor (DSP), a network processor, a graphics processor, a network processor, a communications processor, a cryptographic processor, a co-processor, an embedded processor, or any other type of logic capable of processing instructions.

Processor **401**, which may be a low power multi-core processor socket such as an ultra-low voltage processor, may act as a main processing unit and central hub for communication with the various components of the system. Such processor can be implemented as a system on chip (SoC). Processor **401** is configured to execute instructions for performing the operations discussed herein. System **400** may further include a graphics interface that communicates with optional graphics subsystem **404**, which may include a display controller, a graphics processor, and/or a display device.

Processor **401** may communicate with memory **403**, which in one embodiment can be implemented via multiple memory devices to provide for a given amount of system

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memory. Memory **403** may include one or more volatile storage (or memory) devices such as random access memory (RAM), dynamic RAM (DRAM), synchronous DRAM (SDRAM), static RAM (SRAM), or other types of storage devices. Memory **403** may store information including sequences of instructions that are executed by processor **401**, or any other device. For example, executable code and/or data of a variety of operating systems, device drivers, firmware (e.g., input output basic system or BIOS), and/or applications can be loaded in memory **403** and executed by processor **401**. An operating system can be any kind of operating systems, such as, for example, Windows® operating system from Microsoft®, Mac OS®/iOS® from Apple, Android® from Google®, Linux®, Unix®, or other real-time or embedded operating systems such as VxWorks.

System **400** may further include IO devices such as devices (e.g., **405**, **406**, **407**, **408**) including network interface device(s) **405**, optional input device(s) **406**, and other optional IO device(s) **407**. Network interface device(s) **405** may include a wireless transceiver and/or a network interface card (NIC). The wireless transceiver may be a WiFi transceiver, an infrared transceiver, a Bluetooth transceiver, a WiMax transceiver, a wireless cellular telephony transceiver, a satellite transceiver (e.g., a global positioning system (GPS) transceiver), or other radio frequency (RF) transceivers, or a combination thereof. The NIC may be an Ethernet card.

Input device(s) **406** may include a mouse, a touch pad, a touch sensitive screen (which may be integrated with a display device of optional graphics subsystem **404**), a pointer device such as a stylus, and/or a keyboard (e.g., physical keyboard or a virtual keyboard displayed as part of a touch sensitive screen). For example, input device(s) **406** may include a touch screen controller coupled to a touch screen. The touch screen and touch screen controller can, for example, detect contact and movement or break thereof using any of a plurality of touch sensitivity technologies, including but not limited to capacitive, resistive, infrared, and surface acoustic wave technologies, as well as other proximity sensor arrays or other elements for determining one or more points of contact with the touch screen.

IO devices **407** may include an audio device. An audio device may include a speaker and/or a microphone to facilitate voice-enabled functions, such as voice recognition, voice replication, digital recording, and/or telephony functions. Other IO devices **407** may further include universal serial bus (USB) port(s), parallel port(s), serial port(s), a printer, a network interface, a bus bridge (e.g., a PCI-PCI bridge), sensor(s) (e.g., a motion sensor such as an accelerometer, gyroscope, a magnetometer, a light sensor, compass, a proximity sensor, etc.), or a combination thereof. IO device(s) **407** may further include an imaging processing subsystem (e.g., a camera), which may include an optical sensor, such as a charged coupled device (CCD) or a complementary metal-oxide semiconductor (CMOS) optical sensor, utilized to facilitate camera functions, such as recording photographs and video clips. Certain sensors may be coupled to interconnect **410** via a sensor hub (not shown), while other devices such as a keyboard or thermal sensor may be controlled by an embedded controller (not shown), dependent upon the specific configuration or design of system **400**.

To provide for persistent storage of information such as data, applications, one or more operating systems and so forth, a mass storage (not shown) may also couple to processor **401**. In various embodiments, to enable a thinner and lighter system design as well as to improve system

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responsiveness, this mass storage may be implemented via a solid state device (SSD). However, in other embodiments, the mass storage may primarily be implemented using a hard disk drive (HDD) with a smaller amount of SSD storage to act as a SSD cache to enable non-volatile storage of context state and other such information during power down events so that a fast power up can occur on re-initiation of system activities. Also a flash device may be coupled to processor **401**, e.g., via a serial peripheral interface (SPI). This flash device may provide for non-volatile storage of system software, including a basic input/output software (BIOS) as well as other firmware of the system.

Storage device **408** may include computer-readable storage medium **409** (also known as a machine-readable storage medium or a computer-readable medium) on which is stored one or more sets of instructions or software (e.g., processing module, unit, and/or processing module/unit/logic **428**) embodying any one or more of the methodologies or functions described herein. Processing module/unit/logic **428** may represent any of the components described above. Processing module/unit/logic **428** may also reside, completely or at least partially, within memory **403** and/or within processor **401** during execution thereof by system **400**, memory **403** and processor **401** also constituting machine-accessible storage media. Processing module/unit/logic **428** may further be transmitted or received over a network via network interface device(s) **405**.

Computer-readable storage medium **409** may also be used to store some software functionalities described above persistently. While computer-readable storage medium **409** is shown in an exemplary embodiment to be a single medium, the term “computer-readable storage medium” should be taken to include a single medium or multiple media (e.g., a centralized or distributed database, and/or associated caches and servers) that store the one or more sets of instructions. The terms “computer-readable storage medium” shall also be taken to include any medium that is capable of storing or encoding a set of instructions for execution by the machine and that cause the machine to perform any one or more of the methodologies of embodiments disclosed herein. The term “computer-readable storage medium” shall accordingly be taken to include, but not be limited to, solid-state memories, and optical and magnetic media, or any other non-transitory machine-readable medium.

Processing module/unit/logic **428**, components and other features described herein can be implemented as discrete hardware components or integrated in the functionality of hardware components such as ASICs, FPGAs, DSPs or similar devices. In addition, processing module/unit/logic **428** can be implemented as firmware or functional circuitry within hardware devices. Further, processing module/unit/logic **428** can be implemented in any combination hardware devices and software components.

Note that while system **400** is illustrated with various components of a data processing system, it is not intended to represent any particular architecture or manner of inter-connecting the components; as such details are not germane to embodiments disclosed herein. It will also be appreciated that network computers, handheld computers, mobile phones, servers, and/or other data processing systems which have fewer components or perhaps more components may also be used with embodiments disclosed herein.

Some portions of the preceding detailed descriptions have been presented in terms of algorithms and symbolic representations of operations on data bits within a computer memory. These algorithmic descriptions and representations are the ways used by those skilled in the data processing arts

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to most effectively convey the substance of their work to others skilled in the art. An algorithm is here, and generally, conceived to be a self-consistent sequence of operations leading to a desired result. The operations are those requiring physical manipulations of physical quantities.

It should be borne in mind, however, that all of these and similar terms are to be associated with the appropriate physical quantities and are merely convenient labels applied to these quantities. Unless specifically stated otherwise as apparent from the above discussion, it is appreciated that throughout the description, discussions utilizing terms such as those set forth in the claims below, refer to the action and processes of a computer system, or similar electronic computing device, that manipulates and transforms data represented as physical (electronic) quantities within the computer system's registers and memories into other data similarly represented as physical quantities within the computer system memories or registers or other such information storage, transmission or display devices.

Embodiments disclosed herein also relate to an apparatus for performing the operations herein. Such a computer program is stored in a non-transitory computer readable medium. A non-transitory machine-readable medium includes any mechanism for storing information in a form readable by a machine (e.g., a computer). For example, a machine-readable (e.g., computer-readable) medium includes a machine (e.g., a computer) readable storage medium (e.g., read only memory ("ROM"), random access memory ("RAM"), magnetic disk storage media, optical storage media, flash memory devices).

The processes or methods depicted in the preceding figures may be performed by processing logic that comprises hardware (e.g. circuitry, dedicated logic, etc.), software (e.g., embodied on a non-transitory computer readable medium), or a combination of both. Although the processes or methods are described above in terms of some sequential operations, it should be appreciated that some of the operations described may be performed in a different order. Moreover, some operations may be performed in parallel rather than sequentially.

Embodiments disclosed herein are not described with reference to any particular programming language. It will be appreciated that a variety of programming languages may be used to implement the teachings of embodiments disclosed herein.

In the foregoing specification, embodiments have been described with reference to specific exemplary embodiments thereof. It will be evident that various modifications may be made thereto without departing from the broader spirit and scope of the embodiments disclosed herein as set forth in the following claims. The specification and drawings are, accordingly, to be regarded in an illustrative sense rather than a restrictive sense.

What is claimed is:

1. A method for managing operation of devices in a distributed system, the method comprising:

obtaining, by a recipient and from an initiator, a request for performance of an activity;
making a first determination regarding whether the request is verifiable using local verification data maintained by the recipient by at least:
identifying a signature from the request; and
identifying that the local verification data lacks a certificate chain delegating authority for the activity to an entity that established the certificate, the certifi-

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cate chain also being usable to authenticate the signature with respect to a root of trust for the recipient;

in a first instance of the first determination where the request is not verifiable:

identifying a remote verifier for the request;
requesting that the remote verifier attempt to verify the request by at least:

generating a challenge for the remote verifier, the challenge requiring cryptographic data usable to establish trust in the entity; and

sending the challenge to the remote verifier;

making a second determination regarding whether the remote verifier successfully verified the request; and

in a first instance of the second determination where the remote verifier successfully verified the request;

performing, by the initiator, the activity.

2. The method of claim 1, further comprising:

in a second instance of the first determination where the request is verifiable:

performing, by the initiator, the activity.

3. The method of claim 2, further comprising:

in a second instance of the second determination where the remote cannot verify the request:

rejecting the request for performance of the activity.

4. The method of claim 1, wherein making the second determination comprises:

obtaining a challenge response to the challenge; and

in an instance of the obtaining when the challenge response comprises a signature of a trusted entity and an endorsement of authority of the entity to initiate the activity;

concluding that the remote verifier verified the request.

5. The method of claim 1, wherein requesting that the remote verifier attempt to verify the request further comprises:

identifying that the remote verifier is unreachable, wherein the challenge is sent to the remote verifier after the remote verifier becomes reachable.

6. The method of claim 1, wherein the request does not comprise verification data for the request, the verification data being usable to verify that the initiator has authority to initiate the activity by the recipient.

7. The method of claim 6, further comprising:

obtaining, by the recipient and from a second initiator, a second request for performance of an activity, the second request comprising second verification data for the second request, the second verification data being usable to verify that the second initiator has authority to initiate the activity by the recipient.

8. The method of claim 7, wherein the initiator is operably connected to the recipient via a limited channel, the limited channel preventing the verification data from being part of the request.

9. A non-transitory machine-readable medium having instructions stored therein, which when executed by at least one processor, cause a system to perform system first operations for managing operation of devices in a distributed system, the system first operations comprising:

obtaining, by a recipient and from an initiator, a request for performance of an activity;

making a first determination regarding whether the request is verifiable using local verification data maintained by the recipient by at least:

identifying a signature from the request; and

identifying that the local verification data lacks a certificate chain delegating authority for the activity to

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an entity that established the certificate, the certificate chain also being usable to authenticate the signature with respect to a root of trust for the recipient;

in a first instance of the first determination where the request is not verifiable:

identifying a remote verifier for the request;

requesting that the remote verifier attempt to verify the request by at least:

generating a challenge for the remote verifier, the challenge requiring cryptographic data usable to establish trust in the entity; and

sending the challenge to the remote verifier;

making a second determination regarding whether the remote verifier successfully verified the request; and

in a first instance of the second determination where the remote verifier successfully verified the request;

performing, by the initiator, the activity.

10. The non-transitory machine-readable medium of claim 9, wherein the operations further comprise:

in a second instance of the first determination where the request is verifiable:

performing, by the initiator, the activity.

11. The non-transitory machine-readable medium of claim 10, wherein the operations further comprise:

in a second instance of the second determination where the remote cannot verify the request:

rejecting the request for performance of the activity.

12. The non-transitory machine-readable medium of claim 9, wherein making the second determination comprises:

obtaining a challenge response to the challenge; and

in an instance of the obtaining when the challenge response comprises a signature of a trusted entity and an endorsement of authority of the entity to initiate the activity:

concluding that the remote verifier verified the request.

13. The non-transitory machine-readable medium of claim 9, wherein requesting that the remote verifier attempt to verify the request further comprises:

identifying that the remote verifier is unreachable, wherein the challenge is sent to the remote verifier after the remote verifier becomes reachable.

14. The non-transitory machine-readable medium of claim 9, wherein the request does not comprise verification data for the request, the verification data being usable to verify that the initiator has authority to initiate the activity by the recipient.

15. A data processing system, comprising:

a processor; and

a memory coupled to the processor to store instructions, which when executed by the processor, cause the data processing system to perform operations for managing operation of devices in a distributed system, the operations comprising:

obtaining, by the data processing system and from an initiator, a request for performance of an activity, the data processing system being associated with a recipient of the request;

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making a first determination regarding whether the request is verifiable using local verification data maintained by the data processing system by at least:

identifying a signature from the request; and

identifying that the local verification data lacks a certificate chain delegating authority for the activity to an entity that established the certificate, the certificate chain also being usable to authenticate the signature with respect to a root of trust for the recipient;

in a first instance of the first determination where the request is not verifiable:

identifying a remote verifier for the request;

requesting that the remote verifier attempt to verify the request by at least:

generating a challenge for the remote verifier, the challenge requiring cryptographic data usable to establish trust in the entity; and

sending the challenge to the remote verifier;

making a second determination regarding whether the remote verifier successfully verified the request; and

in a first instance of the second determination where the remote verifier successfully verified the request;

performing, by the initiator, the activity.

16. The data processing system of claim 15, wherein the operations further comprise:

in a second instance of the first determination where the request is verifiable:

performing, by the initiator, the activity.

17. The data processing system of claim 16, wherein the operations further comprise:

in a second instance of the second determination where the remote cannot verify the request:

rejecting the request for performance of the activity.

18. The data processing system of claim 15, wherein making the second determination comprises:

obtaining a challenge response to the challenge; and

in an instance of the obtaining when the challenge response comprises a signature of a trusted entity and an endorsement of authority of the entity to initiate the activity:

concluding that the remote verifier verified the request.

19. The data processing system of claim 15, wherein requesting that the remote verifier attempt to verify the request further comprises:

identifying that the remote verifier is unreachable, wherein the challenge is sent to the remote verifier after the remote verifier becomes reachable.

20. The data processing system of claim 15, wherein the request does not comprise verification data for the request, the verification data being usable to verify that the initiator has authority to initiate the activity by the recipient.

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